Journal of proteome research

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Penicillin–Streptomycin Treatment Rewires Core Metabolic and Ribosomal Programs in HepG2 Cells

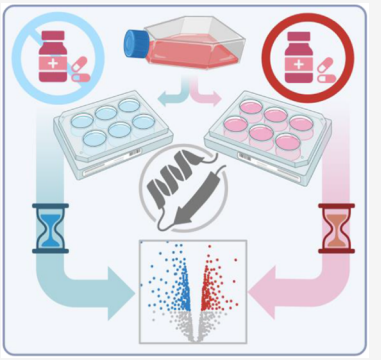
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Cite This: *J. Proteome Res.* 2026, 25, 1082–1094

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ABSTRACT: Antibiotics are routinely added to mammalian cell culture media to prevent bacterial growth. However, the use of antibiotics in a cell culture can confound downstream experimental results. While genomic and transcriptomic differences between cell cultures treated with and without antibiotics are well-documented, far fewer, if any, comprehensive proteomic comparisons on the use of antibiotics in cell culture have been performed. Here, we present a study on the proteome-wide differences of culturing HepG2 cells in antibiotic (i.e., penicillin/streptomycin) and nonantibiotic-containing media. Using a longitudinal and crossover treatment study design, we analyzed 119 samples across nine passages and four conditions. On average, 9,374 proteins were detected per sample, and we identified 383 proteins that were differentially abundant between conditions. These changes included ribosomal and mitochondrial proteins, demonstrating that off-target effects of antibiotics on mammalian cells occur at the protein level. Linear mixed-effect modeling suggested that the proteomic impact of antibiotic treatment is strongest in the first passage after treatment and stabilizes after approximately three passages. Furthermore, initiating antibiotic treatment induced a greater number of differentially abundant proteins than discontinuing treatment. Lastly, we compared our results to existing literature on the use of common antibiotics in mammalian cell culture. We identified proteins and pathways conserved across studies, omics layers, and cell types. We hope that this detailed proteomic survey of the ubiquitous pencillin–streptyomcin-treated HepG2 *in vitro* model will aid researchers in comparing cross-study or cross-condition results from antibiotic-treated mammalian cells and inform appropriate experimental designs for the use of antibiotics in cell culture.



KEYWORDS: *proteomics, antibiotics, HepG2, mitochondria, metabolism, ribosome, lipid metabolism*

INTRODUCTION

Mammalian cell culture media is commonly supplemented with antibiotics for their bacteriostatic or bactericidal effects.[1] Regardless of active bacterial contamination, antibiotics (e.g., penicillin/streptomycin; PenStrep) are often added by default to lower the risk of future contamination. Although chronic treatment of cell cultures with antibiotics is generally accepted practice, others have advocated instead for strict aseptic techniques and the use of antibiotics only when contamination is suspected.[2−4] Chronic reliance on antimicrobials may lead to poor adherence to aseptic protocols, allowing latent contamination to go unnoticed and increasing the risk of drug-resistant microbes.

Studies on the unintentional effects of antibiotic supplementation on cell properties have emerged as an alternative reason to forego such treatment.[3−9] Antibiotic exposure is linked to changes in cell growth, proliferation, differentiation, as well as cytotoxic and apoptosis-inducing properties.[4,9] At high concentrations, they can inhibit protein synthesis and affect enzymatic activity as early as 4 h after treatment.[5,7] At the genomic level, Ryu et al.[3] used RNA-seq to identify over 200 differentially expressed genes in HepG2 cells grown with versus without PenStrep.[3]

While this evidence suggests that the proteome would also be affected, few studies have examined the impact of antibiotic treatment on cell cultures at the protein level. Of these studies, the panels of measured proteins ranged from dozens to hundreds,[6] which is relatively small in comparison to recent proteomic methods capable of identifying over 10,000 proteins.[10] Given the ubiquity with which cell culture media are supplemented with antibiotics for *in vitro* proteomic experiments, a broader investigation into the proteome-wide effects is warranted. Antibiotics may dysregulate the proteins or pathways of interest in an *in vitro* model, leading to follow-up or translational studies that are confounded by the extent and type of antibiotic treatment. Conflicting studies also hinder integrative approaches to multiomics, as data from varying antibiotic treatments are not considered. In summary, while the choice of antibiotic supplementation in cell culture is ostensibly

Received: September 26, 2025
Revised: December 4, 2025
Accepted: December 17, 2025
Published: December 24, 2025



ACS Publications

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https://doi.org/10.1021/acs.jproteome.5c00934
J. Proteome Res. 2026, 25, 1082−1094

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a simple experimental variable, the downstream effects on biologically relevant changes to the proteome remain largely unexplored.

Pen/Strep is one of the most common antibiotic treatments for chronic supplementation in mammalian cell cultures. Their mode of action and effects on bacteria are well understood: penicillin is a β-lactam that inhibits cell-wall synthesis, while streptomycin is an aminoglycoside that targets bacterial ribosomal subunits.[2] They are often used in combination due to their synergistic properties, e.g., inhibition of cell wall synthesis by penicillin may allow easier access of streptomycin to bacterial ribosomes.[11] However, both penicillin and streptomycin have documented toxic and other off-target effects in eukaryotic cells, the exact mechanisms of which are still being elucidated.[12] For example, aminoglycosides have been recently shown to bind to human rRNA,[13] and as many as 75% of mammalian mitochondrial proteins are of bacterial evolutionary origin.[14] Thus, antibiotics likely perturb mitochondrial and ribosomal proteins in eukaryotes. As such, dysregulation of protein synthesis and metabolic pathways resulting from antibiotic treatment alone may be confounded by other biological variables.[8]

Here, we measured proteome remodeling induced by penicillin and streptomycin in HepG2 cells. HepG2 cells were chosen due to their ease of handling, rapid proliferation, and suitability for drug response studies.[15,16] We employed a longitudinal study design with treatment crossover to ensure robust results. In total, we detected approximately 10,250 proteins from 119 samples across nine passages and four treatment groups. As many as 139 proteins with Benjamini–Hochberg adjusted p-values <0.01, or 383 proteins at adj. p-values <0.1 (to be consistent with transcriptomics from Ryu et al.[3]), were differentially abundant between the four conditions. These results were compared in a survey of relevant literature, revealing several conserved proteins and pathways that are consistently affected across omics layers, such as proteostasis, transmembrane transport, ribosome biogenesis, and lipid metabolism. Notably, we observed a discordance between transcriptomic and proteomic changes, suggesting that the effects of antibiotic exposure may be context-specific and modulated by factors, such as passage number, cell type, dosage, and treatment history. We hope that these results and the corresponding data set can provide the proteomics community with a comprehensive resource for HepG2 cells undergoing penicillin–streptomycin treatment. Through the identification of potentially conserved biological responses across diverse literature studies, future work can focus on the applicability to a deeper understanding of the systematic effects of antibiotic supplementation in cell culture and help guide researchers in accounting for these effects in their *in vitro* experimental designs.

METHODS

Materials

The following materials were obtained from ThermoFisher Scientific (Waltham, MA): Dulbecco's Modified Eagle Medium (DMEM; 11995-065); phosphate-buffered saline (PBS; 14190144); dialyzed fetal bovine serum (FBS; A3382001); penicillin–streptomycin (PenStrep; 15-140-122); radioimmunoprecipitation assay buffer (RIPA; AAJ62524AE); ammonium bicarbonate (AC393210010); 0.25% Trypsin-EDTA (25-200-056); 1 M Tris-HCl pH 8 (15568025); LC-MS grade 99% formic acid (A117–50); LC-MS grade 0.1% formic acid in water (v/v) (LS118–1). The following materials were obtained from Millipore Sigma (Burlington, MA): Roche cOmplete protease inhibitor cocktail mini tablets (11836153001); HPLC grade ethanol (59828-1L); anhydrous calcium chloride ($CaCl_2$, C5670-100G); HPLC-grade acetonitrile (34851); Pierce HeLa protein digest standards (88329); dithiothreitol (DTT); and 3-indoleacetic acid (IAA). Sequencing-grade modified trypsin (V5113) was purchased from Promega (Madison, WI). Silica magnetic beads (786-916) were purchased from G-Biosciences (St. Louis, MO).

Cell Culture

HepG2 cells were a gift from Shelly Lu's lab. The HepG2 cells were cultured in high-glucose DMEM with 10% fetal bovine serum (FBS) and antibiotics (1% PenStrep; 100 U/mL penicillin, 100 μg/mL streptomycin) for five passages to ensure the absence of latent microbial contamination. All incubations occurred at 37 °C and 5% CO_2. At passage 22, one confluent T-75 flask of HepG2 cells was split into two six-well tissue-culture-treated plates. Each well of each plate contained a seeding density of roughly 0.2×10^6 cells. One plate contained cells in high-glucose DMEM with 10% FBS and 1% PenStrep (antibiotic-containing media; A), and the other plate contained cells in high-glucose DMEM with 10% FBS (no antibiotics; N). Media changes were performed approximately every 2 days. After approximately 72 h, roughly equal confluency of >80% ($\sim 1.2 \times 10^6$ cells) was confirmed across plates using an inverted microscope. One well of each plate was used to seed the six-well plate for the following passage, e.g., one well of A_1 was used to seed the entire plate of A_2. This was done by placing 1 mL of 0.25% trypsin-EDTA in the well, incubating for 3 min, diluting with serum-free DMEM, centrifuging at 1,000 rpm for 5 min, aspirating, and resuspending the pellet in 12 mL of the corresponding treatment media. The remaining five wells were rinsed with PBS and lysed using 125 μL of RIPA with a dissolved protease inhibitor cocktail tablet (1 tablet/10 mL) and then placed on ice for 5 min. Each well was then scraped and transferred to Eppendorf tubes. The tubes were centrifuged at 14,000 rpm for 15 min and sonicated for 30 s, and the supernatant was transferred to a new Eppendorf tube. Samples were stored at −80 °C before sample preparation. This process was repeated for passages 1–5. On passage 6, to split into crossover conditions, two wells were harvested per plate, while the remaining four were lysed as before. One well was resuspended in antibiotic-containing media, and the other in nonantibiotic-containing media. This resulted in four total plates from passage six onward: A_6, A_5N_1, N_6, N_5A_1. The harvest and lysis protocol was then performed as described for passages 1–5 for the remaining passages up until passage 9. All cultures were checked roughly every 24 h for any evidence of microbial contamination using an inverted microscope. We accounted for the position of the well during cell culture by passaging from the bottom row of the six well plates. One well (or two wells, in the case of passage 5) was chosen randomly from the bottom three rows to use to seed for the next passage. The remaining wells were analyzed. Any well-specific effects (temperature, evaporation, etc., due to the position on the six well plate) should only be seen between the middle and outer edges of the 6-well plates and should thus be averaged out by randomly sampling the bottom row.

Sample Preparation

All samples were prepared on the same day using an automated single-pot, solid-phase enhanced sample preparation protocol

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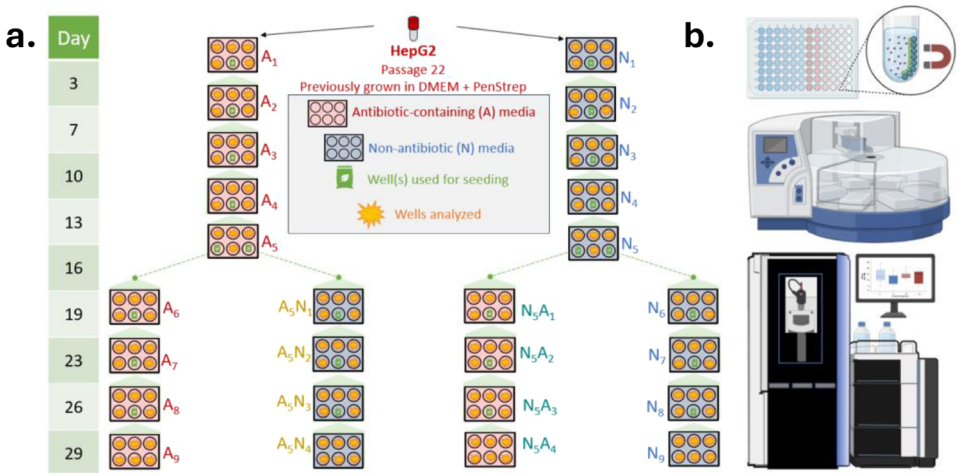


Figure 1. Experimental overview. **a)** HepG2 cells were grown in Dulbecco's Modified Eagle Medium (DMEM) containing penicillin/streptomycin (PenStrep). At passage 22, a confluent T-75 flask was split into two six-well plates and grown in the same media, except that one plate contained antibiotics (A) and the other did not (N). Each plate was grown under the same conditions for five passages; the subscript represents the number of passages after splitting the cells into each treatment group (i.e., A_{1-5}, N_{1-5}). After five passages, A_5 and N_5 were split again, each into two 6-well plates. The treatment either remained constant and grew for an additional four passages (i.e., A_{6-9}, N_{6-9}) or was swapped to the opposite condition (i.e., N_5A_{1-4}, A_5N_{1-4}). For each passage, one or two wells were used to seed the subsequent passage plate, while the remaining wells were analyzed using the proteomics workflow. **b)** The proteomics workflow involved using single-pot, solid-phase enhanced sample preparation, followed by liquid chromatography–mass spectrometry with an Orbitrap Astral.

(SP3)[17] using a KingFisher Flex (ThermoFisher). Samples were thawed, and 5.8 μL of lysate was transferred into each well (~33 μg protein/well). Including quality control samples, approximately 160 samples were prepared across two 96-well plates. The quality controls included a pooled sample (5 μL of all samples mixed together) and a blank (containing only RIPA buffer; no cell lysate). Samples were reduced by incubation with DTT for 30 min at RT, followed by alkylation with IAA for 30 min at RT in the dark. Next, 6.6 μL of paramagnetic silica beads were added to achieve a 1:10 ratio of protein: beads (w/w). Beads were washed with Milli-Q water using the KingFisher flex prior to sample addition. Next, 105 μL of acetonitrile (LC-MS grade) was added to each well to achieve a final concentration of 70% (v/v) acetonitrile in each well. The samples were incubated at RT for 18 min with gentle agitation. Next, the samples with beads containing precipitated protein were washed twice in 500 μL of 80% ethanol, followed by two washes in 500 μL of 100% acetonitrile. The beads were then transferred into a 50 mM solution of Tris-HCl containing 10 mM $CaCl_2$ and 1:25 protein: trypsin (w/w). Samples were covered and incubated with gentle agitation at 37 °C overnight. The next day, the beads were removed from the samples, and the samples were diluted with 5.5 μL of 10% formic acid followed by 109 μL of 0.1% formic acid. The plates were then centrifuged and transferred to a new 96-well plate for analysis.

LC-MS

Samples were organized into a blocking pattern containing one sample from each treatment and passage. The injection order was randomized within each block. A blank and pooled sample was run before and after each block. The performance of the system was tested by using injections of solvent blanks and HeLa protein digest standards. An injection volume of 10 μL was used. A Vanquish Neo LC was used with a PepSep C18 15 cm × 150 μm, 1.5 μm at 55 °C (Bruker Corporation, Billerica, MA). A 30 min chromatography gradient was used, using 0.1% formic acid in water (solvent A) and 80% acetonitrile with 0.1% formic acid in water (solvent B). The starting composition of solvent B was 4%, increasing to 9% at 2 min, 25% at 18 min, and 35% at 27 min before column wash and equilibration. A pump curve value of 5 and a flow rate of 1.2 μL/min were used for all gradient steps. A Thermo Orbitrap Astral instrument was used for mass spectrometry. Data were acquired using data-independent acquisition with the Orbitrap resolution set to 240,000 and a 3 m/z isolation window spanning 380–980 m/z.

Data Analysis

DIANN 1.9 was used to quantitate proteins from the raw data in the library-free mode. A human reference proteome .fasta file (downloaded January 27, 2025) containing canonical proteins and isoforms (83,062 proteins and 20,562 genes) was used to generate an *in silico* library. Three missed cleavages were allowed. All other settings were default. The output pg_matrix from DIANN was analyzed in Jupyter notebooks using Python version 3.12.2, NumPy version 1.26.4, pandas version 2.2.2, and anndata version 0.10.9. Plots were made using matplotlib 3.8.4 and seaborn 0.13.2. Preprocessing, imputation, and principal components analyses (PCA) were done using scanpy version 1.10.2 and scikit-learn version 1.4.2. Prior to PCA analysis, the data were scaled across both the samples and proteins. After removing blanks, quality control samples, and 7 sample outliers from the data set, any proteins present in less than 50% of samples were removed. This resulted in 9,518 proteins out of 10,255 (93%) proteins. All remaining 119 samples contained at least 8,924 proteins (<7% missingness), up to 9,374 proteins in the highest sample (average ± standard deviation of 9,210 ± 106 proteins). Samples were log-transformed and then normalized by the total protein quantity present prior to imputation of missing values using a k-nearest neighbors imputer with five nearest neighbors. Each sample was then annotated by plate (i.e., 1–2), replicate (i.e., 1–5), passage (i.e., 1–9), and condition (i.e., A, N, AN, or NA). Statistical tests such as t tests were done using the scipy stats package version 1.13.1. Multiple test corrections were performed using the Benjamini-Hochberg correction, and the adjusted p-value significance threshold was set to a $p \leq 0.01$ cutoff, unless otherwise noted. Linear mixed

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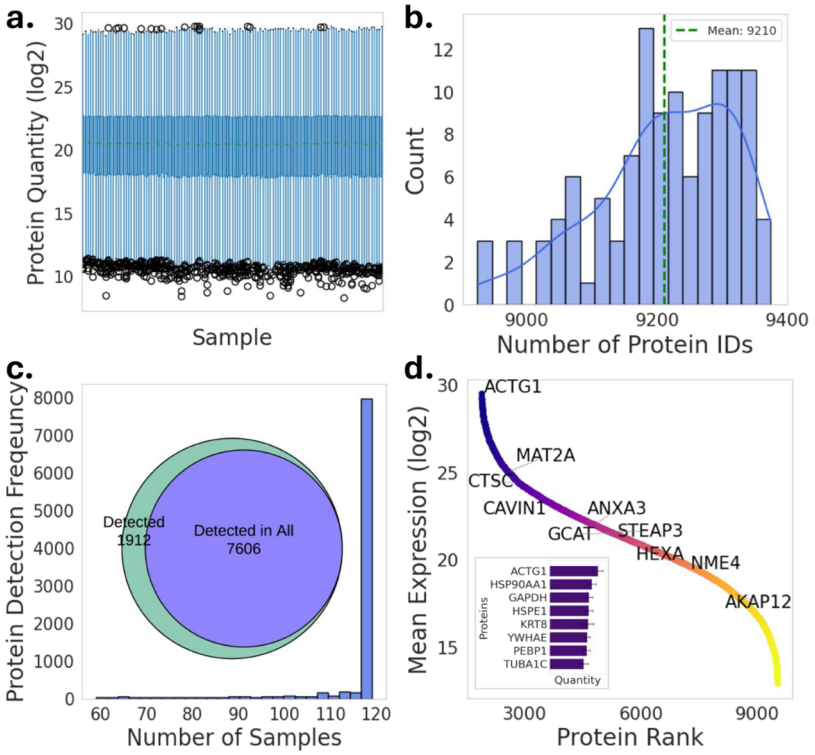


Figure 2. Protein quantification overview. **a)** Boxplot of protein quantities for all samples. **b)** Histogram of the number of detected proteins across all samples. **c)** Histogram of the number of samples in which each protein was detected. Inset: Euler diagram of all detected proteins versus the proteins detected in all samples. **d)** Average quantities of proteins across all samples. Inset: top eight most abundant proteins. All values are prior to imputation.

effect models utilized statsmodels 0.14.2. The gseapy package, version 1.1.4, was used for gene set enrichment analyses. All STRING diagrams were created using string-db.org (version 12.0) with the full network type, with evidence-based edges, all active interaction sources, minimum low confidence interaction scores (0.150), and no interactors.

RESULTS AND DISCUSSION

Experimental Design

The experimental overview is shown in Figure 1. Briefly, two six-well plates of HepG2 cells (seeded at the same density and from the same passage) were grown in parallel in antibiotic (A) and nonantibiotic (N) media (i.e., 1% PenStrep; see Methods). On each passage, at least one well was used to seed the following passage, while the remaining wells were washed, lysed, and frozen until the day of analysis. Throughout, cells were seeded at approximately equal densities (~0.2×10^6 cells per well) and harvested at equal confluency (>80% or ~1.2×10^6 cells), as confirmed visually using an inverted microscope. In the fifth passage of the parallel A and N treatments, each condition was further split into two subgroups (a continued treatment group or a reverse treatment group) for four additional passages. Continued treatment refers to the use of the same media conditions, e.g., A_{1-9}, N_{1-9}. Reverse, or crossover, treatment refers to switching the media to the opposing treatment; for example, cells grown in antibiotics for five passages were then grown in nonantibiotic-containing media for four passages (A_5N_{1-4}). This experimental design enabled us to continue monitoring a single condition for an extended period (29 days; 9 passages: A_{1-9}, N_{1-9}), while also replicating treatment transitions in both directions (A_5N_{1-4}, N_5A_{1-4}). On the same experimental day, all lysates were then thawed and underwent solid-phase enhanced sample preparation (SP3).[17] Samples were then frozen until analyzed by liquid chromatography-tandem mass spectrometry with data-independent acquisition (see Methods).

This study design has the following advantages. First, it enables the longitudinal tracking of proteomic changes across nine sequential passages, in contrast to prior studies that typically examined only single-passage time points or short-term antibiotic exposure (Table S2). By crossing the conditions (i.e., crossover design), we increase the robustness and statistical power of our study by capturing both onset and withdrawal effects of antibiotic treatment within the same experimental framework. Changes in protein abundance that remain consistent across conditions and passages are likely to reflect generalizable biological responses rather than transient effects due to passage-specific variation or genetic drift.[18]

Global Protein Differences

The protein quantities were consistent across all samples (Figure 2a). Approximately 9,200 proteins were detected on average (Figure 2b). A total of 10,255 proteins were detected, and on average, each protein was detected in 108/119 samples (91%; Figure 2c). Remarkably, 7,606 proteins were detected in all samples. The most abundant proteins included those related to cytoskeletal organization, stress response, and metabolism (Figure 2d). Taken together, these results are expected, given the consistent handling and treatment of the cells, aside from the intended variation of antibiotic supplementation.

We then analyzed the variations in total protein quantities across plates, treatment conditions, and passages (Figures S1, S2 and Supplementary Note S1). All variations were consistent across these experimental variables. A slight decrease in protein content was noted for specific wells of passages 5 and 6 due to

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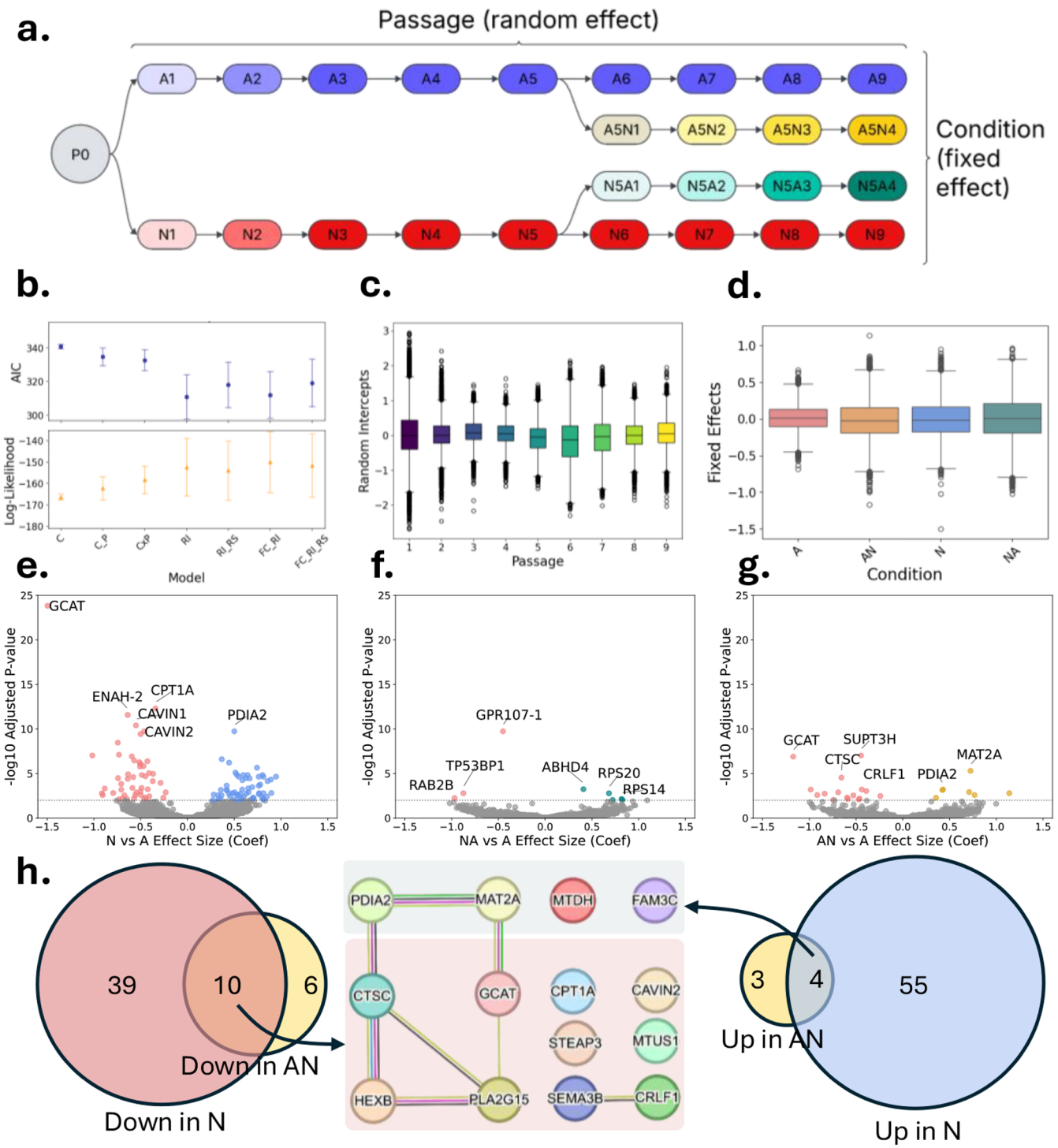


Figure 3. Linear mixed effect model reveals significant protein differences across conditions and passages. a) Flowchart distinguishing fixed versus random effects in the context of treatment condition versus passage. b) Model comparison metrics for model selection, where AIC represents Akaike information criterion. C, C_P, and CxP present ordinary least-squares models with covariates of condition, condition + passage, and condition*passage, respectively. RI and RI_RS represent random intercept and random intercept + random slope linear mixed effect models, respectively, where passage is the random effect. FC_RI and FC_RI_RS are the same as the latter models, except that the condition has been added as a fixed effect. c) Box plot of random intercepts for the FC_RI model. d) Box plot of the fixed effects for the FC_RI model. e–g) Volcano plots for each condition, where effect is given by the FC_RI fixed effect coefficient. An adjusted p-value using Benjamini–Hochberg correction was set to a $p \leq 0.01$ cutoff. h) Euler diagrams for the proteins differentially abundant in each condition (left/right). STRING diagram for proteins consistently differentially abundant across conditions (middle).

variations in confluency, but this did not significantly alter any downstream results (see Figures S3–S5).

Linear Mixed-Effect Modeling

Previous studies comparing antibiotic treatments on cultured cells have used t tests or similar analyses to compare the log-fold

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J. Proteome Res. 2026, 25, 1082–1094

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changes of genes or proteins across treatment (A) versus control (N) at a single time point or passage.[3,8] For direct comparison to these studies, we provide a discussion of *t* test results in Figures S6−S9, Supplementary Note S2, and Table S1. Interestingly, we found that moving from antibiotic-free cultures to antibiotic-containing cultures (N → A) consistently resulted in more differentially abundant proteins than moving from antibiotic-containing cultures to antibiotic-free cultures (A → N).

Given the longitudinal, multipassage nature of our data set, these pairwise time-point comparisons risk obscuring meaningful biological variation with passage-dependent variation. To address this, we utilized a linear mixed-effect model (LME). LMEs are useful for grouped data sets to incorporate both fixed effects (i.e., parameters expected to vary systematically across an entire population) and random effects (i.e., related to individualized experimental units or subjects).[19] In our design, we considered the two primary sources of variation (treatment condition and passage) as fixed and random effects, respectively (Figure 3a). Further discussion on the motivation and selection of LMEs compared to other linear models or statistical analyses is provided in the Supporting Information (Supplementary Note S3).

Differential Expression

To determine the most appropriate linear model for our data, we evaluated multiple models using standard model selection criteria (Figure 3b). The best-performing model was an LME with treatment condition as a fixed effect and passage as a random intercept (FC_RI; see Supplementary Note S3 and Figures S10−S12). The distributions of the random intercepts demonstrated that passage-by-passage variation was greatest in the first passage directly after a condition crossover (passages 1, 6), but stabilizes after roughly three passages (passages 3, 8) (Figure 3c). These rapid changes in many individualized protein coefficients reflect transient HepG2 proteome remodeling following changes in antibiotic (penicillin/streptomycin) exposure.

Using the FC_RI model, the fixed effect coefficients (Figure 3d) can then be used to perform differential expression analysis of proteins across the conditions (Figure 3e−g). Here, we use A as the reference, as antibiotic-containing media is commonly used by default, as was the case in this study, and compare against N (no antibiotics), NA (crossover from N to A), and AN (crossover from A to N), respectively. In Figure 3e, positive coefficients are higher in N, and negative coefficients are higher in A. The *y*-axis represents adjusted *p*-values using the Benjamini−Hochberg procedure with a significance threshold of ≤0.01, unless otherwise noted.

Using this threshold, we found 108 proteins differentially abundant in N versus A (59 up, 49 down), 8 proteins differentially abundant in NA versus A (5 up, 3 down), and 23 proteins differentially abundant in AN versus A (7 up, 16 down) (Figure 3h). Across these three comparisons, this resulted in 139 unique differentially abundant proteins (71 up, 68 down, relative to A). Of these, FAM3C, MAT2A, MTDH, and PDIA2 were increased in both N and AN (i.e., consistently higher abundance when not exposed to antibiotics).

Meanwhile, CAVIN2, CPT1A, CRLF1, CTSC, GCAT, HEXB, MTUS1, PLA2G15, SEMA3B, and STEAP3 were decreased in both N and AN (i.e., consistently higher abundance when exposed to antibiotics; Figure 3h, center). The full list of all proteins differentially abundant across conditions is provided in Table S3 (XLSX).

For brevity, we focus on the 14 select proteins differentially abundant across multiple comparisons, which is a benefit of our crossover design (Figure 3h, center). The differentially abundant protein with the lowest adjusted *p*-value and greatest fixed effect coefficient was glycine C-acetyltransferase (GCAT). This protein was consistently more abundant in the presence of antibiotics (penicillin/streptomycin). GCAT is a mitochondrial enzyme involved in generating glycine as part of the threonine degradation pathway. Glycine is one of the three amino acids required to synthesize the endogenous tripeptide glutathione, which plays a crucial role in the oxidative stress response, metabolic detoxification, and immune system regulation.[20] Thus, the increase in GCAT in response to antibiotics may be implicated in oxidative stress protection mechanisms. Few studies have investigated GCAT, but some evidence suggests that this enzyme, along with others related to glycine metabolism, is implicated in metabolic regulation and aging.[21−23]

Along this same pathway related to one-carbon metabolism, methionine adenosyltransferase 2A (MAT2A) catalyzes the synthesis of S-adenosylmethionine (SAM) from methionine. SAM is required for the trans-sulfuration pathway to generate cysteine, which is then required for glutathione production. Glycine, and potentially GCAT activity, feeds into the methionine cycle.[24] However, unlike GCAT, MAT2A was found to decrease in the presence of antibiotics. While GCAT localizes to the mitochondria, MAT2A resides primarily in the nucleoplasm and cytoplasm and is more active in extrahepatic tissues. Due to the production of SAM, MAT2A is implicated in methylation, which, in turn, affects gene expression and post-translational modifications. Its presence in the nucleus and cytoplasm could also indicate its effects on ribosomal biogenesis and function. Regardless, these results suggest a regulatory effect of the glycine- and glutathione-related metabolism.

Changes in protein folding and lipid oxidation pathways parallel this shift in the redox-related metabolism. Protein disulfide-isomerase A2 (PDIA2) was found to decrease in the presence of antibiotics. PDIA2 is located in the endoplasmic reticulum and is believed to function as a protein folding chaperone. Carnitine palmitoyltransferase 1A (CPT1A), found on the outer mitochondrial membrane, is a rate-limiting enzyme involved in fatty acid beta-oxidation. CPT1A was more abundant in the presence of antibiotics. While we hypothesized changes to mitochondrial proteins would occur, this specific alteration of a crucial lipid metabolism protein was unexpected.

Several membrane-associated proteins were also found to be differentially abundant. Caveolae-associated proteins (CAVINs), which contribute to membrane curvature and scaffolding, were decreased in response to antibiotics. These proteins have various roles associated with everything from the cytoskeleton to ribosomes, transcription, lipid metabolism, hypoxia, and cell signaling.[25] Six-transmembrane epithelial antigen of prostate 3 (STEAP3) is another membrane protein that is also decreased in AN and N, with multiple roles in liver health and disease.[26]

Other robustly affected proteins include cytokine receptor-like factor 1 (CRLF1) and semaphorin-3B (SEMA3B), which are secreted proteins that regulate immune responses and apoptosis.[27,28] The remaining proteins are associated with hepatocellular carcinoma, cell proliferation, and tumor growth, including *MTUS1 (microtubule-associated scaffold protein 1), metadherin (MTDH), and FAM3C (family with sequence similarity 3, metabolism-regulating signaling molecule C)*. For example, MTUS1 is thought to be a tumor suppressor such that

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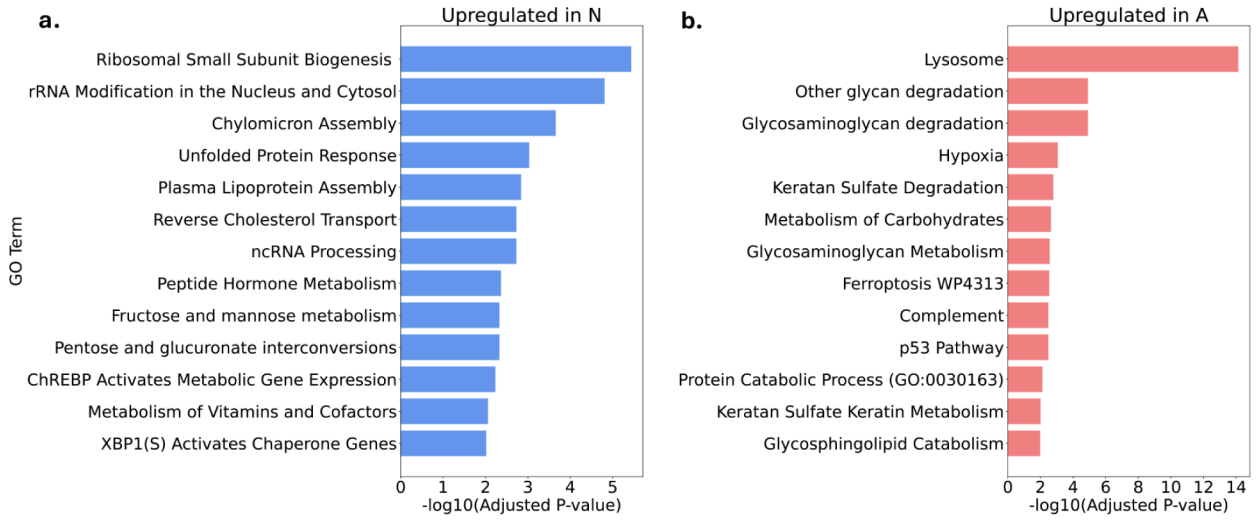


Figure 4. GO figures for terms upregulated in samples a) without or b) with antibiotics. Significant proteins were those with adjusted p-values ≤ 0.01. Redundant or irrelevant terms were manually removed. The unedited figures can be found in Figures S13,14. The gene sets used were: MSigDB_Hallmark_2020, GO_Biological_Process_2023, Reactome_Pathways_2024, WikiPathways_2024_Human, and KEGG_2021_Human (accessed June 5, 2025).

downregulation is associated with poor prognosis.[29] Meanwhile, MTDH[30] and FAM3C[31] are known to be upregulated in various cancers, including hepatocellular carcinoma. In our data, MTUS1 was consistently increased with antibiotics, while MTDH and FAM3C were consistently decreased with antibiotics. These and other differences reflect existing literature detailing the benefits and risks of antibiotics in cancer treatment and progression.[32]

Pathway Analysis

To better understand the biological pathways in HepG2 cells affected by penicillin−streptomycin treatment, we performed gene ontology (GO) enrichment analyses on the differentially abundant proteins (Figure 4). To assess general pathway trends, we grouped proteins that were more abundant in either A or NA, or less abundant in AN, as broadly increased under antibiotic exposure (a total of 65 unique proteins). Conversely, proteins more abundant in N or AN, or less abundant in NA, were grouped as generally increased in nonantibiotic conditions (a total of 60 unique proteins). The relevant filtered pathway results are listed in Figure 4. The unfiltered results are shown in Figures S13−S14.

Our original hypotheses were that antibiotics cause off-target effects on mitochondrial and ribosomal proteins. These plots reflect such results. Antibiotics may induce mitochondrial stress, resulting in alterations in metabolic pathways. This explains why terms like hypoxia are enriched in A, and various metabolic processes (e.g., lipids, carbohydrates; Carbohydrate Response Element Binding Protein (ChREBP1 pathway)) are altered across both conditions (Figure 4). This is consistent with known effects of aminoglycosides like streptomycin, which, in addition to targeting bacterial ribosomes, can bind eukaryotic mitochondrial and cytosolic ribosomes.[12] We indeed found enrichment of ribosome biogenesis and rRNA terms in nonantibiotic-containing media (Figure 4a). Thus, mitochondria (via altered metabolism) and ribosomal effects are reflected at the protein level.

Several additional pathway enrichments provided insight into the broader cellular consequences of antibiotic exposure. For example, the increase in unfolded protein response, p53, complement, and ferroptosis pathways, which are enriched in antibiotic-conditions (Figure 4b), may explain the observations regarding antibiotics' effects on cell proliferation, cytotoxicity, and cancer.[32]

Literature Comparison

To contextualize our findings, we reviewed several relevant studies detailing the effects of antibiotics on mammalian cell cultures (Table S2). From this, Table 1 highlights proteins and protein families that were differentially abundant in our study and in at least one other relevant publication.

In particular, our study follows a similar design to that of Ryu et al.[3] Ryu et al.[3] used HepG2 cells grown from passage 37 under the same antibiotic- and nonantibiotic media used here, except in a single 6-well plate (3 replicates) for two passages (21 days).[3] Using RNA-seq and ChIP-seq, the authors found 209 genes that were differentially expressed between A and N. While our study differs in sample size and exposure time, the results of their transcriptomics studies allow for comparison with our proteomic results (Figure 5 and Table 1). Of the 9,964 genes in the Ryu et al.[3] data set annotated with UniProt IDs, we extracted 206 differentially expressed genes.

Overall, we found substantial overlap in coverage of our proteomic data set with the transcriptomic data set of Ryu et al.;[3] after filtering for redundant isoforms, the overlap was nearly 80% (7,503) (Figure 5a). We then compared the log2 fold changes reported by Ryu et al.[3] (from their second passage) with our log2 fold changes from our second passage, focusing only on treatment groups N versus A. The results were not correlated between studies (Figure 5b). To test whether this discrepancy was due to differences in exposure duration rather than passage number, we performed an additional comparison using our sixth passage (19 days), which more closely matched the 21-day exposure in Ryu et al.[3] This analysis yielded a small, but still statistically insignificant, positive correlation (r = 0.02, Figure S15). This lack of correlation may be expected, as transcriptomics and proteomics are known to be poorly correlated,[33,34] especially under oxidative stress.[35] Even within the report of Ryu et al.,[3] the correlation between ChIP-seq and RNA-seq was as low as 0.15, underscoring the complexity of

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Table 1. Proteins of Interest Were Found to Be Differentially Abundant in This Study and at Least One Other Relevant Study[a]

	Reference						
Biological Relevance	This work	Ryu et al.[3]	Mathieson et al.[6]	Elliot et al.[8]	Relier et al.[40]	Kalghatgi et al.[39]	Han et al.[41]
Calcium homeostasis	RGN	RGN					
Cell cycle/signaling	CDK16						CDK2
	PPP1R2	PPP2CB					
	TOP2B	TOP1					
	IGFBP1	IGFBP1,3					
Cytoskeleton	ACTG2, ACTR1A, PHACTR2, ENAH		ACT*	ACTB			ACTB
	KRT19,23,80	KRT23					
	AKAP12	AKAP12					
Glycolytic and ATP metabolic processes	**NME**3,4	NME1					
	PFKP			PFKM			
	PKLR		PKLR				
	AK1	AK2					
	ATP6 V0A1	ATP2B2					
Immune response	**ANXA**3,4				ANXA5		
Lipid metabolism	PLIN2	PLIN2					
	PLA2G15	PLA2G12B					
	ACSL3	**ACS**S3,S5,L1,L5,M3					
	PECR	PECR					
	ABAT	ABAT					
Oxidative stress response	**ALDH**1A1, L2, 4A1,18A1				ALDH*		
	GPX2					GPX1	
Protein folding, chaperones, and proteostasis	LRRC20	LRRC59					
	HSPB1, 90B1	**HSP**E1,A5,9,H1	**HSP**10,70,90				
	BAG2	BAG2					
	HYOU1		HYOU1	HIF1a			
	UBAC1, UBFD1	UBASH3B, UBE2Q1	UBA1				UBB
	NEDD4L	NEDD9	NEDD8				
	VCP		VCP				
	RNF149	RNF10					
Ribosome biogenesis	**UTP**4,6,15,14A	**UTP**15,14A					
	RPF2	RPF2					
	WDR3,36,43	WDR77					
RNA processing	PCBP2						PCBP2
	DDX3,17,27,47,52,54,56						DDX5
	RBM22,28	RBM24					
Translation	**EIF**6,2AK4,4H	**EIF**2B2,3J					
	SRPRB	SRPRB					
Transmembrane transport	**SLC**3A2,39A4, 0,14,O4C1	**SLC**6A14,20A1,7A11, 1A4,35B1,22A7,6A4, 39A4,38A4		**SLC**2A1,3			
Xenobiotic response	ABCB1	**ABC**B1,C6P1					
Transcription regulation	ZFAND3, ZNF281	ZFP36L					

[a]Underlined proteins/genes denote an exact match. Bolded families are prefixes for multiple proteins listed. An asterisk denotes an unspecified protein match.

gene regulatory responses to antibiotics at different biological levels.

We then compared our differentially abundant proteins from the LME model with those defined as differentially expressed by Ryu et al.[3] (Figure 5c). To match their adjusted *p*-value threshold, we reduced our adjusted *p*-value cutoff to <0.1 and identified a total of 383 differentially abundant proteins (208 upregulated and 175 downregulated). Ryu et al.[3] identified a total of 209 differentially expressed genes (157 upregulated and 52 downregulated). Of these, 14 exact matches (by UniProt ID) were found (Figure 5d). Five proteins were significantly upregulated in A in both data sets, and three proteins were significantly upregulated in N in both data sets. However, six proteins were downregulated in Ryu et al.[3] but upregulated in our data. No proteins downregulated in our data were reported as upregulated in Ryu et al.[3]

A reason for this mismatch could be due to differences in passage numbers, passage-associated drift, experimental design or interlaboratory (rather than matched-sample) differences, or the well-known discordance between proteomes and transcriptomes. Our work utilized HepG2 cells at passage 22 with a longitudinal crossover design, whereas Ryu et al.[3] employed cells at passage 37, which were collected after two passages, for two treatment conditions. Noted (epi)genetic differences occur in immortalized cell lines across passages in relation to their primary line, especially at high passage numbers.[18] However, relative differences between closely related passages of established cell lines are the more commonly encountered and

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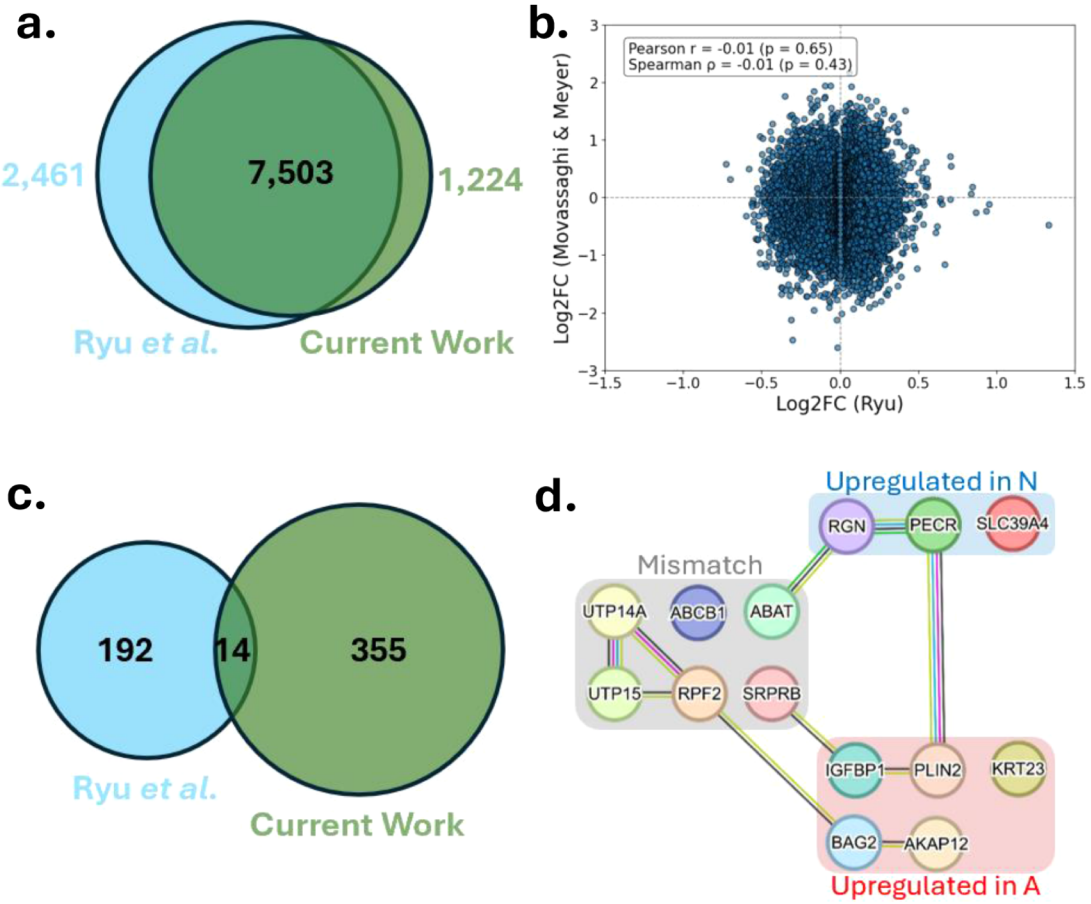


Figure 5. Literature comparison. **a)** Overlap of proteins detected in this work versus gene names (with UniProt IDs) reported by Ryu et al.[3] **b)** Fold-change (log2 scale) correlation between this work and Ryu et al.[3] Both fold changes are from the second passage only. **c)** Overlap of significant proteins. A threshold of $p \leq 0.1$ was used to match the report by Ryu et al.[3] **d)** STRING diagram of the overlapping proteins from **c**.

experimentally realistic scenarios. Furthermore, it was unclear whether the previous passages in Ryu et al.[3] utilized antibiotic- or nonantibiotic-containing media. This shift in treatment baseline can result in contradictory findings, as suggested by Elliot et al.[8]

To broaden the search for similarities, we examined matches within the same isoform or protein family (Table 1), as well as similarly enriched pathways across a survey of relevant studies. For example, we found CPT1A to be differentially abundant, while Ryu et al.[3] found dysregulated mitochondrial carnitine shuttle pathways. CPT1A is the rate-limiting enzyme within the carnitine shuttle pathway, essential for fatty acid oxidation, which β-lactams are known to inhibit.[36] Similarly, Ryu et al.[3] reported dysregulation of forkhead box (FOX) transcription factors, which can bind with XBP1, an unfolded protein response (UPR) pathway enriched in our data set.[37] Other UPR pathways include eukaryotic translation initiation factors (EIFs), found in both our data set and Ryu et al.[3] The ubiquitin/proteasome protein degradation pathway is also activated as proteins accumulate, and protein translocation in the endoplasmic reticulum (ER) is mediated by ribosomal-binding SEC proteins (SEC11, 31, and 61 were all increased in N in our data).[38] This can increase the level of ER stress, leading to lipid accumulation in hepatocytes. Excess lipids can be transiently stored as lipid droplets. Lipid droplets can then undergo lysosome-dependent autophagy, which is reflected by our findings of differential abundance in alpha-beta hydrolase domain (ABHD), RAB, heat-shock (HSP), and perilipin (PLIN) proteins. Accordingly, we also identified differentially abundant lysosomal proteins involved in lipid metabolism, including hexosaminidase B (HEXB), phospholipase A2 (PLA2G15), and cathepsin C (CTSC). These pathways explain the robust changes in proteins related to lipid metabolism, ER stress, chaperones, and protein turnover across studies (Table 1 and Figure 6). They may be a downstream effect of altered ribosomal or mitochondrial function.

Other, smaller-scale studies using various antibiotics and cell lines also provide insight into conserved biological changes. Llobet et al.[9] demonstrated that the use of penicillin, streptomycin, amphotericin (AmB), and gentamicin influences the differentiation of human adipose tissue-derived stem cells into adipocytes.[9] They found a 40% increase in reactive oxygen species (ROS) production in the presence of antibiotics, as measured by the rise in hydrogen peroxide production. When treated with the antioxidant *N*-acetyl-L-cysteine (NAC) in addition to antibiotics, intracellular lipid droplets and leptin staining decreased by up to 70%, compared to increases of up to 36% and 290%, respectively, in PenStrep. Interestingly, in contrast with our and Ryu et al.'s[3] results, authors did not find significant effects on mitochondrial enzymes using colorimetric assays.

Kalghatgi et al.[39] explored aminoglycosides and β-lactams in relation to mitochondrial dysfunction and oxidative damage *in vitro* (using MCF10A cells) and *in vivo* (in mice).[39] The authors

https://doi.org/10.1021/acs.jproteome.5c00934
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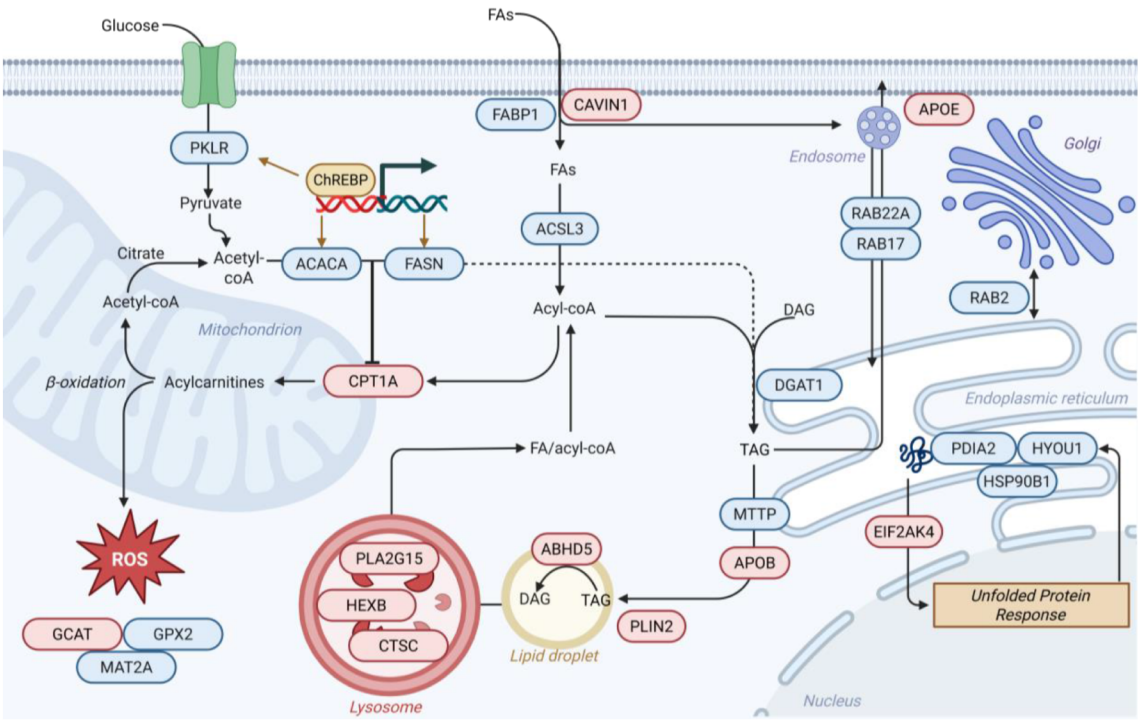


Figure 6. Possible dysregulated protein pathways involved in lipid transport and metabolism for HepG2 cells treated with or without antibiotics. Red UniProt IDs represent proteins upregulated in antibiotics, and blue represents proteins upregulated in nonantibiotics. All proteins shown were differentially abundant in this work.

found that clinically relevant treatments caused mitochondrial dysfunction and the overproduction of ROS. For example, glutathione peroxidase (GPX1), among other antioxidant defense genes, was upregulated by up to 10-fold after prolonged antibiotic treatment. Interestingly, these effects were also alleviated by the addition of NAC. Of these, in our work, we found that GPX2 was higher in cells without antibiotics.

Relier et al.[40] examined the effect of PenStrep on the phenotypes of cancer cells.[40] They found that these antibiotics inhibited the sphere-forming ability in six cancer cell lines across three different tissue types. Staining for annexin V (ANXA5) revealed no significant differences after PenStrep treatment, while our results showed increased ANXA isoforms in the antibiotic-containing samples. Meanwhile, staining for aldehyde dehydrogenase (ALDH) showed a substantial decrease in PenStrep, which is in agreement with the ALDH isoforms detected in our data set.

Elliot et al.[8] investigated the effects of gentamicin in three mammary cell lines.[8] This study showed a significant upregulation of hypoxia-inducible factor 1 (HIF1a), phosphofructokinase (PFK-M), lactate dehydrogenase (LDH-A), and glucose transporters SLC2A1 and SLC2A3 in gentamicin-treated samples relative to controls. Gentamicin was also found to significantly increase mitochondrial superoxide production within 24 h. These results are in general agreement that solute carrier family (SLC) proteins as well as glycolysis and ROS pathways have conserved effects for antibiotics beyond PenStrep.

Mathieson et al.[6] used MCF-7 breast cancer cells with PenStrep, AmB, and control.[6] AmB is a commonly used antifungal agent and is often combined with PenStrep to form an antimicrobial cocktail (i.e., "anti-anti"). They demonstrated that no differences in contamination, dead cells, or cell cycle phase would account for between-culture variations in confluence. Instead, they reported widespread decreases in chaperone, cytoskeletal, and ER stress-related proteins in PenStrep-treated cells. In our data set, hypoxia upregulated 1 protein (HYOU1), HSP90, valosin-containing protein (VCP), and pyruvate kinase (PKLR) were similarly decreased, further supporting a conserved stress-response proteome.

Han et al.[41] reported the dose-dependent effects of gentamicin, streptomycin, and penicillin on mouse blastocysts using RNA-seq[41] For the lowest doses tested, roughly 1,800 genes were significantly differentially expressed. For penicillin and streptomycin specifically, the top KEGG pathways affected included p53, RNA splicing and transport, ubiquitin-mediated proteolysis, cholesterol and lipid metabolism, and glycan biosynthesis, which align with pathway results from our work (Figure 4).

Based on the above literature, summarized in Table 1, a consistent alteration in lipid metabolism was observed. Other conserved effects across studies include protein turnover, cytoskeletal maintenance, oxidative stress, ribosome biogenesis, and membrane transporters. A list of possible protein pathways dysregulated in lipid metabolism is provided in Figure 6. All proteins/pathways shown were differentially abundant in our data.

For example, the carbohydrate response element-binding protein (ChREBP) pathway was enriched with N (Figure 4). ChREBP is a transcription factor that regulates glycolytic and lipogenic proteins such as pyruvate kinase (PKLR), acetyl-CoA carboxylase alpha (ACACA), and fatty acid synthase (FASN),[37] all of which are more abundant in N. The latter proteins are involved in *de novo* lipogenesis, where ACACA produces malonyl-CoA, followed by the rate-limiting production of palmitate by FASN. Malonyl-CoA is a known inhibitor of CPT1A, preventing long-chain fatty acid oxidation.[37] Meanwhile, in A, CPT1A and thus fatty acid oxidation are

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upregulated, generating ROS. The liver is a key regulator of glutathione, which protects against ROS, and also has roles in detoxication of xenobiotics.[20] This may account for the dysregulation of various proteins involved in glutathione synthesis via amino acid metabolism (GCAT, MAT2A) or antioxidant response (GPX2), the latter being ferroptosis-related (a process upregulated in part A). However, glutathione synthesis is rate-limited by the availability of cysteine and the activity of glutamate-cysteine ligase. The latter accounts for MAT2A, which is involved in the methionine cycle to produce cysteine. Some evidence suggests that glutathione synthesis may be jointly regulated by both glycine and cysteine.[42]

The liver is highly metabolic and principally utilizes fatty acids. Free fatty acids enter liver cells via diffusion or active transport by proteins like CD36 and are eliminated via oxidation or exocytosis of lipoproteins. Meanwhile, cholesterol and lipoproteins enter the cytosol through receptor-mediated endocytosis. These processes are heavily regulated to maintain a steady state and prevent accumulation. In HepG2 cells, fatty acid-binding protein 1 (FABP1) and CAVIN1 are involved in the transport of fatty acids.[37,38] FABP1 is also found in the cytosol, where it binds various lipid substrates. Acyl-CoA synthetase long-chain (ACSL) enzymes then produce fatty acyl-CoA. ACSL1 is found on the outer mitochondrial membrane and interacts with CPT1A in hepatocytes. Both ACSL1 and ACSL5 were identified by Ryu et al.[3] and are known to be highly expressed in the liver,[37] while we identified the upregulated ACSL3 isoform in the absence of antibiotics. ACSL3 is expressed on lipid droplets and is involved in de novo lipogenesis. Meanwhile, acyl-CoA thioesterase 2 (ACOT2) catalyzes the reverse reaction of ACSL isoforms, leading to lipid catabolism, and was upregulated in N. The coordinated regulation of ACOT and ACSL remains under study.[38]

During triglyceride synthesis, diacylglycerol acyltransferase (DGAT) transforms diacylglycerols (DAGs) to triacylglycerols (TAGs) in the ER, which can then form lipid droplets.[38] These droplets can then undergo autophagy with a lysosome or be secreted as a very-low-density lipoprotein (VLDL). In lipid droplets, the perilipin (PLIN) superfamily and microsomal triglyceride transfer protein (MTTP) incorporate TAGs into apolipoprotein B (APOB), which moves from the ER to Golgi to form VLDL.[38] The transport of VLDL also accounts for the differential abundance of several vesicle trafficking proteins (RAB2B,17,22A and SEC11C,31A,61B) found to be increased in N.

Overall, the lipid metabolism signaling in our data appears mixed, with some proteins indicating more catabolism and others indicating more storage. A possible source of error is differences in cell density.[43] However, several precautions were taken to avoid such effects, including matched cell seeding densities, visual confirmation of matched confluency throughout the experiment, and the use of linear mixed-effect models to identify consistent effects across passages. We note some limitations of this work. First, this study focuses on HepG2 cells and penicillin/streptomycin treatment as a case study system. Second, the transcriptome work presented here was from the literature, not matched samples. Still, we maintain that the use of antibiotics in cell cultures for proteome experiments is an overlooked source of variation. For example, antibiotics are commonly employed as only acute treatments. Our work provides guidance that reconditioning for at least three passages is needed for the HepG2-PenStrep system. Similar unaccounted-for variation and effects on reproducibility may be encountered when donating cell lines between laboratories or integrating data across studies. Future work may focus on elucidating the precise mechanisms underlying these bidirectional regulatory differences.

CONCLUSIONS

Despite its ubiquity, we demonstrate that supplementing HepG2 cell culture media with antibiotics (penicillin and streptomycin) yields a range of ribosomal and mitochondrial variations in the proteome. The breadth of our results enabled comparisons across protein, transcript, and pathway-level analyses. While we observe some conserved agreements related mainly to lipid metabolism, oxidative stress, and protein turnover, the magnitude and direction of many effects appear to be dependent on cell line, culture conditions, concentrations, and time (including passage), as well as the type of antibiotic used. In agreement with previous transcriptomics advice, proteomic studies (especially those involving *in vitro* models of metabolism) should consider the confounding effects of antibiotic supplementation. Although this study utilizes only one mammalian cell line (i.e., HepG2) and one antibiotic combination (i.e., penicillin/streptomycin) as a representative *in vitro* mammalian cell model, this work has broad implications. In particular, for the bulk proteomic analysis of mammalian cells, it is likely that care should be taken when 1) comparing results across studies using antibiotics versus not, or using different types of antibiotics; and 2) if treating acutely with antibiotics, cell cultures should be reconditioned in control media for at least three passages to ensure a more consistent proteomic profile. Future studies should examine additional cell lines and other antibiotics to understand which proteome alterations are common and divergent. Lastly, a potential area for future work is the use of antioxidants, such as *N*-acetyl-L-cysteine, to mitigate off-target effects involving reactive oxygen species, in conjunction with multiple, diverse cell lines and antibiotic treatments to uncover fundamental and conserved changes in cellular processes driven by antibiotics.

ASSOCIATED CONTENT

Data Availability Statement

All raw data and DIANN output files can be accessed through MassIVE under data set identifier MSV000098243 (doi:10.25345/C59P2WK1T). All analysis code can be accessed at https://github.com/xomicsdatascience/hepg2_abc.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jproteome.5c00934.

Exploration of variation in quantified proteins (Note 1); global differences across conditions (Figure S1); principal component scores plot annotated by plate and condition (Figure S2); recreated panels from Figures S1 and S2 with passages 4 and 5 excluded (Figure S3); recreated panels from Figure 3 from the main text with passages 4 and 5 excluded (Figure S4); recreated Figure 4 from the main text with passages 4 and 5 excluded (Figure S5); exploration of data analysis using simple *t* tests (Note 2); volcano plot of all passages of A versus N using a two-sided *t* test (Figure S6); volcano plots of passages 1−9 for comparing conditions N versus A (Figure S7); volcano plots of passages 6−9 comparing crossover conditions (Figure S8); the number of significantly differentially abundant proteins found in each passage/condition

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shown in Figures S7 and S8 (Table S1); the number of differentially expressed proteins by condition and cross-over treatment, stacked by passage (Figure S9); discussion on the motivation and selection of LMEs compared to other linear models or statistical analyses (Note 3); random and effect plots for alternative linear mixed effect models (LME) (Figure S10); recreated Figure 3 (main text) using FC_RI_RS results(Figure S11); recreated Figure 4 (main text) using FC_RI_RS results (Figure S12); summary of relevant studies on effects of antibiotic treatment in mammalian cell culture-(Table S2); unfiltered GO term enrichment results, upregulated in N (Figure S13); unfiltered GO term enrichment results, upregulated in A (Figure S14); comparison of passage 6 of A versus N log2 fold changes with those reported by Ryu et al.[3] (Figure S15) (PDF)

Measured proteins and their coefficients from linear mixed effects modeling with corrected *p*-values for each comparison (Table S3) (XLSX)

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Author Contributions

C.S.M. and J.G.M. conceived the study. C.S.M. performed the experiments and analyzed the data. J.G.M. supervised the project. All authors wrote and approved the manuscript.

Funding

This study was funded by the NIH NIGMS R35GM142502.

Notes

This research did not involve human or animal participants. The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We thank the Shelly Lu lab for gifting us a stock vial of HepG2 cells and for comments on the manuscript. We thank Bora Onat and Jesus de Jose Munoz Estrada for assistance with cell culture and Simion Kreimer for help in operating the Orbitrap Astral. Biorender.com is acknowledged for the generation of the table of contents graphic and Figures 1, 6.

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https://doi.org/10.1021/acs.jproteome.5c00934
J. Proteome Res. 2026, 25, 1082−1094